

lab18(SSU)

November 3, 2025

1 Lab Assignment 18: Confidence Intervals

Helpful Resource:

- [Python Reference](#): Cheat sheet of helpful array & table methods!

Recommended Reading:

- [Estimation](#)

```
[1]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
!pip install py pandoc
!pip install datascience
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

pandoc set to manually installed.

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntaxtex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl

texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexlua53 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

0 upgraded, 54 newly installed, 0 to remove and 41 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1> [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 fonts-lato all 2.0-2.1> [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 poppler-data all 0.4.11-1> [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu/jammy/universe amd64 tex-common all 6.17> [33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 fonts-urw-base35 all 20200910-1> [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu/jammy-updates/main amd64 libgs9-common all 9.55.0~dfsg1-0ubuntu5.13> [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu/jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1> [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 libijs-0.35 amd64 0.35-15build2> [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 libjbig2dec0 amd64 0.19-3build2> [64.7 kB]

Get:10 <http://archive.ubuntu.com/ubuntu/jammy-updates/main amd64 libgs9 amd64 9.55.0~dfsg1-0ubuntu5.13> [5,032 kB]

Get:11 <http://archive.ubuntu.com/ubuntu/jammy-updates/main amd64 libkpathsea6 amd64 2021.20210626.59705-1ubuntu0.2> [60.4 kB]

Get:12 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 libwoff1 amd64 1.0.2-1build4> [45.2 kB]

Get:13 <http://archive.ubuntu.com/ubuntu/jammy/universe amd64 dvisvgm amd64 2.13.1-1> [1,221 kB]

Get:14 <http://archive.ubuntu.com/ubuntu/jammy/universe amd64 fonts-lmodern all 2.004.5-6.1> [4,532 kB]

Get:15 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 fonts-noto-mono all 20201225-1build1> [397 kB]

Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all 20180621-3.1 [10.2 MB]
Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java all 18-1 [4,720 B]
Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-java all 43-1 [10.8 kB]
Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-java all 1.2-2 [60.3 kB]
Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration all 1.18 [5,336 B]
Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64 3.0.2-7ubuntu2.11 [50.1 kB]
Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems all 3.3.5-2ubuntu1.2 [228 kB]
Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1 [5,100 B]
Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7 kB]
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all 0.1.1-2 [12.6 kB]
Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all 0.3.2-1ubuntu0.1 [24.9 kB]
Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynchronet2 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64 2.5.11+ds1-1 [421 kB]
Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64 0.13.72+dfsg.1-1.1 [27.0 kB]
Get:35 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all 1:1.0.5-0ubuntu2 [578 kB]
Get:36 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64 1:7.7+6build2 [94.6 kB]
Get:37 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all 2.004.5-6.1 [9,471 kB]
Get:38 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 preview-latex-style all 12.2-1ubuntu1 [185 kB]
Get:39 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 t1utils amd64 1.41-4build2 [61.3 kB]

```

Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 4s (47.2 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 125080 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.

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Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...

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Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../32-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...

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Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.

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Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libjbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...

```



```

Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

```

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_openc1.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

Processing triggers for tex-common (6.17) ...

Running updmap-sys. This may take some time... done.

Running mktexlsr /var/lib/texmf ... done.

Building format(s) --all.

This may take some time... done.

Collecting py pandoc

Downloading py pandoc-1.15-py3-none-any.whl.metadata (16 kB)

Downloading py pandoc-1.15-py3-none-any.whl (21 kB)

Installing collected packages: py pandoc

Successfully installed py pandoc-1.15

Collecting datascience

Downloading datascience-0.18.1-py3-none-any.whl.metadata (910 bytes)

Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from datascience) (0.20.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from datascience) (75.2.0)

Requirement already satisfied: matplotlib>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from datascience) (3.10.0)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datascience) (2.2.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from datascience) (1.16.3)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages

(from datascience) (2.0.2)

Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-packages (from datascience) (7.34.0)

Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (from datascience) (5.24.1)

Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages (from datascience) (0.8.2)

Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (3.1.6)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2.32.4)

Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2025.4.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.4.9)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (25.0)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (2.9.0.post0)

Collecting jedi>=0.16 (from ipython->datascience)

 Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)

Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (4.4.2)

Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.7.5)

Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (5.7.1)

Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)

Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (2.19.2)

Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-

```

packages (from ipython->datascience) (0.2.0)
Requirement already satisfied: matplotlib-inline in
/usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.2.1)
Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (4.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->datascience) (8.5.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience)
(0.8.5)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from
jinja2>=2.9->folium>=0.9.1->datascience) (3.0.3)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience)
(0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-
toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.14)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->folium>=0.9.1->datascience) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2025.10.5)
Downloading datascience-0.18.1-py3-none-any.whl (725 kB)
725.1/725.1 kB
37.6 MB/s eta 0:00:00
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
62.3 MB/s eta 0:00:00
Installing collected packages: jedi, datascience
Successfully installed datascience-0.18.1 jedi-0.19.2

```

```

[2]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive

```

```
drive.mount('/content/drive')

import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')
```

Mounted at /content/drive

```
[3]: # Don't change this cell; just run it.

import numpy as np
from datascience import *

# These lines do some fancy plotting magic.",
import matplotlib
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
import warnings
warnings.simplefilter('ignore', FutureWarning)
```

1.1 1. Thai Restaurants in Berkeley

Jessica and Ciara are trying to see what the best Thai restaurant in Berkeley is. They survey 1,500 UC Berkeley students selected uniformly at random and ask each student which Thai restaurant is the best. (*Note: This data is fabricated for the purposes of this homework.*) The choices of Thai restaurants are [Lucky House](#), [Imm Thai](#), [Thai Temple](#), and [Thai Basil](#). After compiling the results, Jessica and Ciara release the following percentages of votes that each restaurant received, from their sample:

Thai Restaurant	Percentage
Lucky House	8%
Imm Thai	53%
Thai Temple	25%
Thai Basil	14%

These percentages represent a uniform random sample of the population of UC Berkeley students. We will attempt to estimate the corresponding *parameters*, or the percentage of the votes that each restaurant will receive from the population (i.e. all UC Berkeley students). We will use confidence intervals to compute a range of values that reflects the uncertainty of our estimates.

The table `votes` contains the results of Jessica and Ciara's survey.

```
[4]: # Just run this cell
votes = Table.read_table('./DS/votes.csv')
votes
```

```
[4]: Vote
Lucky House
Lucky House
Lucky House
Lucky House
Lucky House
Lucky House
Lucky House
Lucky House
Lucky House
Lucky House
Lucky House
... (1490 rows omitted)
```

Question 1.1. Complete the function `one_resampled_percentage` below. It should return Imm Thai's *percentage* of votes after taking the original table (`tbl`) and performing one bootstrap sample of it. Remember that a percentage is between 0 and 100. **(8 Points)**

Note 1: `tbl` will always be in the same format as `votes`.

Note 2: This function should be completed **without** `.group` or `.pivot`. Using these functions will cause your code to timeout.

Hint: Given a table of votes, how can you figure out what percentage of the votes are for a certain restaurant? **Be sure to use percentages, not proportions, for this question!**

```
[10]: def one_resampled_percentage(tbl):
    bootstrap = tbl.sample()
    imm_votes = bootstrap.where('Vote', are.equal_to('Imm Thai')).num_rows
    total_votes = bootstrap.num_rows
    return (imm_votes / total_votes) * 100

one_resampled_percentage(votes)
```

```
[10]: 55.600000000000001
```

Question 1.2. Complete the `percentages_in_resamples` function such that it simulates and returns an array of **2023 elements**, where each element represents a bootstrapped estimate of the percentage of voters who will vote for Imm Thai. You should use the `one_resampled_percentage` function you wrote above. **(8 Points)**

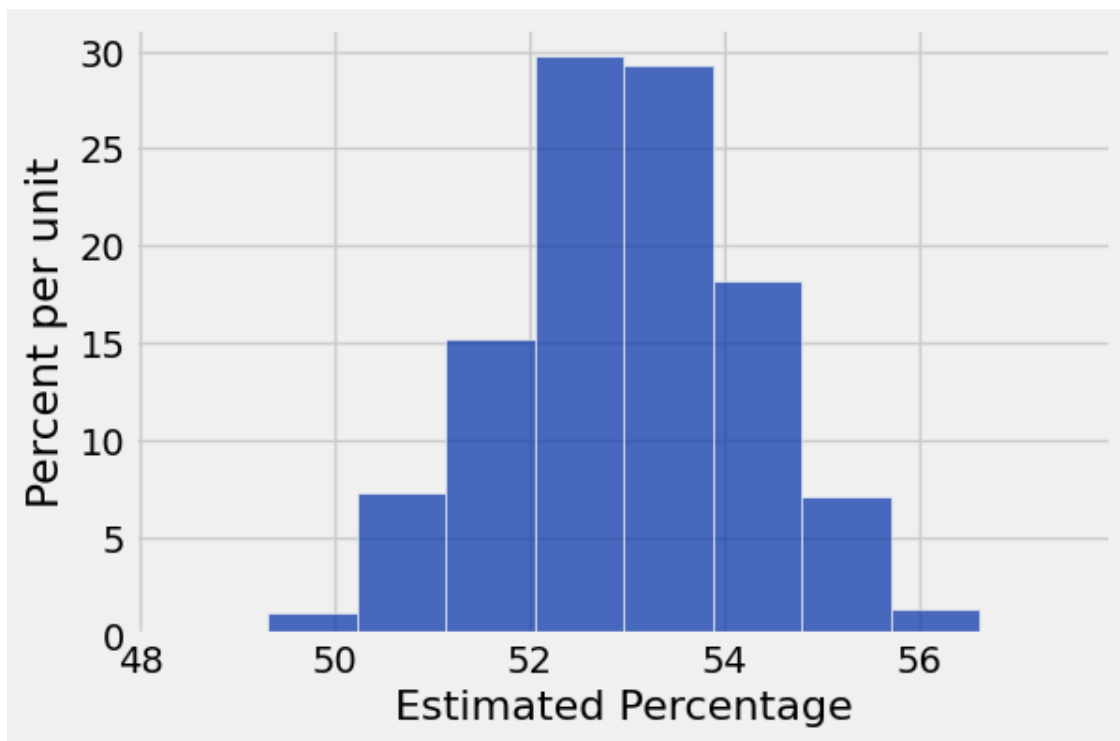
Note: We perform our simulation with **only 2023 trials** in this problem to reduce the runtime, but we should generally use more repetitions.

```
[12]: def percentages_in_resamples():
    percentage_imm = make_array()
    for i in np.arange(2023):
        percentage_imm = np.append(percentage_imm,
    ↪ one_resampled_percentage(votes))
    return percentage_imm
```

In the following cell, we run the function you just defined, `percentages_in_resamples`, and create a histogram of the calculated statistic for the 2023 bootstrap estimates of the percentage of voters who voted for Imm Thai.

Note: This might take a few seconds to run.

```
[13]: resampled_percentages = percentages_in_resamples()
      Table().with_column('Estimated Percentage', resampled_percentages).
      ↪ hist("Estimated Percentage")
```



Question 1.3. Using the array `resampled_percentages`, find the values at the two edges of the middle 95% of the bootstrapped percentage estimates. Compute the lower and upper ends of the interval, named `imm_lower_bound` and `imm_upper_bound` respectively. (8 Points)

Hint: If you are stuck on this question, try looking over [Chapter 13.1](#) of the textbook.

```
[18]: imm_lower_bound = percentile(2.5, resampled_percentages)
      imm_upper_bound = percentile(97.5, resampled_percentages)
      print(f"Bootstrapped 95% confidence interval for the percentage of Imm Thai_
      ↪ voters in the population: [{imm_lower_bound:.2f}, {imm_upper_bound:.2f}]" )
```

Bootstrapped 95% confidence interval for the percentage of Imm Thai voters in the population: [50.47, 55.33]

Question 1.4. The survey results seem to indicate that Imm Thai is beating all the other Thai restaurants among the voters. We would like to use confidence intervals to determine a range of

likely values for Imm Thai's true lead over all the other restaurants combined. The calculation for Imm Thai's lead over Lucky House, Thai Temple, and Thai Basil combined is:

$$\text{Imm Thai's percent of vote} - (100 \text{ percent} - \text{Imm Thai's percent of Vote})$$

Define the function `one_resampled_difference` that returns **exactly one value** of Imm Thai's percentage lead over Lucky House, Thai Temple, and Thai Basil combined from one bootstrap sample of `tbl`. **(8 Points)**

Hint 1: Imm Thai's lead can be negative.

Hint 2: Given a table of votes, how can you figure out what percentage of the votes are for a certain restaurant? **Be sure to use percentages, not proportions, for this question!**

Note: If the skeleton code provided within the function is not helpful for you, feel free to approach the question using your own variables.

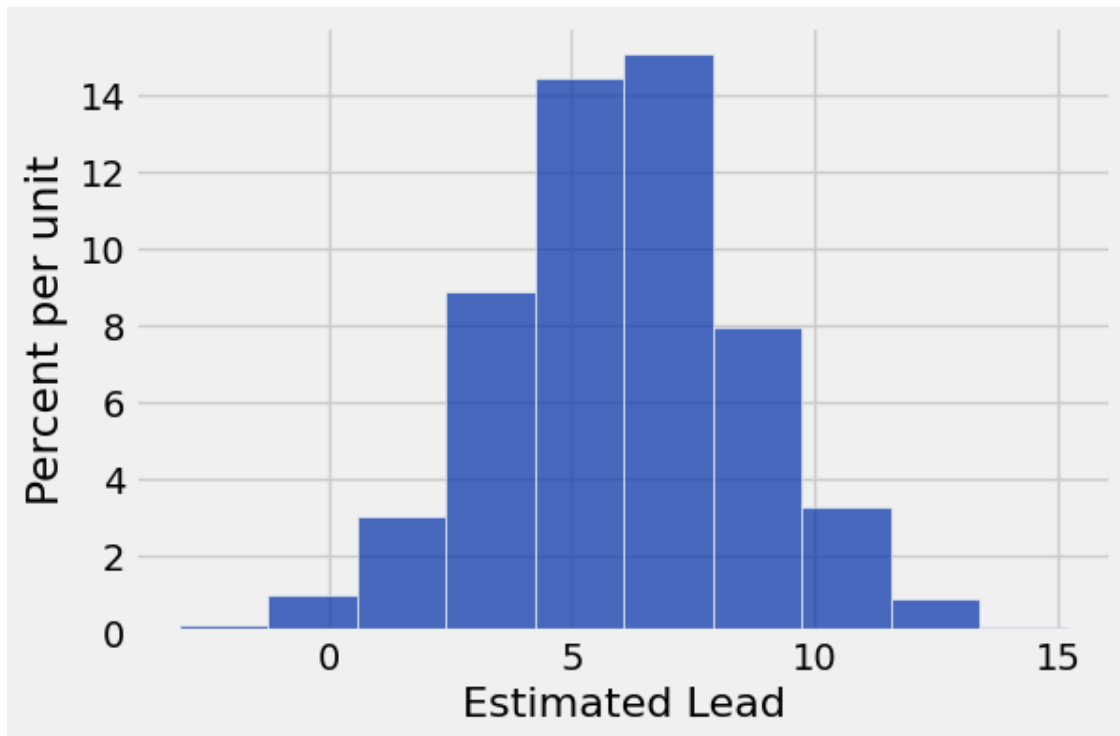
```
[20]: def one_resampled_difference(tbl):
    bootstrap = tbl.sample()
    imm_percentage = (bootstrap.where('Vote', are.equal_to('Imm Thai')).
↳ num_rows / bootstrap.num_rows) * 100
    other_percentage = 100 - imm_percentage
    return imm_percentage - other_percentage
```

Question 1.5. Write a function called `leads_in_resamples` that returns an array of 2023 elements representing the bootstrapped estimates (the result of calling `one_resampled_difference`) of Imm Thai's lead over Lucky House, Thai Temple, and Thai Basil combined. Afterwards, run the cell to plot a histogram of the resulting samples. **(8 Points)**

Hint: If you see an error involving `NoneType`, consider what components a function needs to have!

```
[22]: def leads_in_resamples():
    sampled_leads = make_array()
    for i in np.arange(2023):
        sampled_leads = np.append(sampled_leads,
↳ one_resampled_difference(votes))
    return sampled_leads

sampled_leads = leads_in_resamples()
Table().with_column('Estimated Lead', sampled_leads).hist("Estimated Lead")
```

Question 1.6. Use the simulated data in `sampled_leads` from Question 1.5 to compute an approximate 95% confidence interval for Imm Thai's true lead over Lucky House, Thai Temple, and Thai Basil combined. **(10 Points)**

```
[25]: diff_lower_bound = percentile(2.5, sampled_leads)
diff_upper_bound = percentile(97.5, sampled_leads)
print("Bootstrapped 95% confidence interval for Imm Thai's true lead over Lucky_
↳House, Thai Temple, and Thai Basil combined: [{:f}%, {:f}%]".
↳format(diff_lower_bound, diff_upper_bound))
```

Bootstrapped 95% confidence interval for Imm Thai's true lead over Lucky House, Thai Temple, and Thai Basil combined: [0.933333%, 11.200000%]

1.2 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit. **Please save before exporting!**

```
[ ]: # should change the directory and file name matching to yours
!jupyter nbconvert './DS/lab18(SSU).ipynb' --to pdf
```